

MILOŠ STANOJEVIĆ

LECTURER (ASSISTANT PROFESSOR) AT UCL

SENIOR RESEARCH SCIENTIST AT GOOGLE DEEPMIND

RESEARCH INTERESTS

- Linguistics: Language Universals, Syntactic Formalisms and Incremental Parsing
- Learning: Neural Latent Variable Models and Probabilistic Inference over Structures
- Applications: Machine Translation, Sign Language Generation and LLMs in general

WORK EXPERIENCE

- Sep 2024 – now **Lecturer (~Assistant Professor) in Computational Linguistics**
University College London
Linguistics Department, London, UK
- Nov 2021 – now **Senior Research Scientist**
Google DeepMind
Language Team, London, UK
- Nov 2017 –
Oct 2021 **PostDoc / Research Associate**
University of Edinburgh
School of Informatics, Edinburgh, UK
- July 2017 –
Oct 2017 **Internship**
Nuance Communications (now part of Microsoft)
Natural Language and AI group, Sunnyvale, US

EDUCATION

- 2013 – 2017 **Ph.D. Natural Language Processing**
University of Amsterdam, Netherlands
- 2011 – 2013 **Mgr. Mathematical Linguistics**
Charles University in Prague, Czechia
- 2010 – 2011 **M.Sc. Human Language Science and Technology**
University of Malta, Malta
- 2006 – 2010 **B.Sc. Electrical Engineering**
University of Niš, Serbia
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[HTTPS://STANOJEVIC.GITHUB.IO/](https://stanojevic.github.io/)

- arXiv 2026** [Syntactic Belief Update as the Driver of Garden Path Processing Difficulty](#)
arXiv preprint
 Alan Zhou*, **Miloš Stanojević***, John Hale
 *equal contribution
- HSP 2025** [Generalizing active gap-filling](#)
Conference on Human Sentence Processing
 Tim Hunter, **Miloš Stanojević**, Edward Stabler
- COLING 2025** [Extracting structure from an LLM - how to improve on surprisal-based models of Human Language Processing](#)
International Conference on Computational Linguistics
 Daphne Wang, Mehrnoosh Sadrzadeh, **Miloš Stanojević**, Wing-Yee Chow, Richard Breheny
- EMNLP 2024** [Do LLMs learn a true syntactic universal?](#)
Empirical Methods in Natural Language Processing
 John Hale and **Miloš Stanojević**
- ACL 2024** [Multipath parsing in the brain](#)
Annual Meeting of the Association for Computational Linguistics
 Berta Franzluebbers, Donald Dunagan, **Miloš Stanojević**, Jan Buys, John Hale
- CMCL 2024** [How can large language models become more human?](#)
best paper award *Workshop on Cognitive Modeling and Computational Linguistics*
 Daphne Wang, Mehrnoosh Sadrzadeh, **Miloš Stanojević**, Wing-Yee Chow, Richard Breheny
- EMNLP System Demo 2023** [SynJax: Structured Probability Distributions for JAX](#)
Empirical Methods in Natural Language Processing System Demonstration
Miloš Stanojević and Laurent Sartran
- Cognitive Science journal 2023** [Modeling structure-building in the brain with CCG parsing and large language models](#)
top cited *Cognitive Science — journal*
Cognitive Science **Miloš Stanojević**, Jonathan Brennan, Donald Dunagan, Mark Steedman,
paper in 2023 John Hale
- Neurobiology of Language journal 2023** [Neural correlates of long-distance dependencies in Chinese and English](#)
Neurobiology of Language — journal
 Donald Dunagan, **Miloš Stanojević**, Maximin Coavoux, Shulin Zhang, Shohini Bhattasali, Jixing Li, Jonathan Brennan, John Hale



- EMNLP 2022** [Unbiased and Efficient Sampling of Dependency Trees](#)
Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević
- TACL
journal 2022** [Transformer Grammars: Augmenting Transformer Language Models
with Syntactic Inductive Biases at Scale](#)
Transactions of the Association for Computational Linguistics — journal
 Laurent Sartran, Samuel Barrett, Adhiguna Kuncoro,
Miloš Stanojević, Phil Blunsom, Chris Dyer
- Computational
Linguistics
journal 2021** [Formal Basis of a Language Universal](#)
Computational Linguistics — journal
Miloš Stanojević and Mark Steedman
- EMNLP 2021** [A Root of a Problem: Optimizing Single-Root Dependency Parsing](#)
Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević and Shay B. Cohen
- IWCS 2021** [Computing All Quantifier Scopes with CCG](#)
International Conference on Computational Semantics
Miloš Stanojević and Mark Steedman
- CMCL 2021** [Modeling incremental language comprehension in the brain with
Combinatory Categorical Grammar](#)
Workshop on Cognitive Modeling and Computational Linguistics
Miloš Stanojević, Shohini Bhattachali, Donald Dunagan, Luca Campanelli,
 Mark Steedman, Jonathan Brennan, John Hale
- CASE 2021** [Modality and Negation in Event Extraction](#)
*Workshop on Challenges and Applications of Automated Extraction of
Socio-political Events from Text*
 Sander Bijl de Vroe, Liane Guillou, **Miloš Stanojević**, Nick McKenna and
 Mark Steedman
- ACL 2020** [Max-Margin Incremental CCG Parsing](#)
Annual Meeting of the Association for Computational Linguistics
Miloš Stanojević and Mark Steedman
- IWPT 2020** [Span-Based LCFRS-2 Parsing](#)
The 16th International Conference on Parsing Technologies
Miloš Stanojević and Mark Steedman
- CUNY 2020** [Predictive Processing of Coordination in CCG](#)
Conference on Human Sentence Processing
Miloš Stanojević, John Hale and Mark Steedman



- ACL 2019** [Wide-Coverage Neural A* Parsing for Minimalist Grammars](#)
Annual Meeting of the Association for Computational Linguistics
 John Torr, **Miloš Stanojević**, Mark Steedman and Shay B. Cohen
- FG 2019** [On the Computational Complexity of Head Movement and Affix Hopping](#)
Conference on Formal Grammar
Miloš Stanojević
- NAACL 2019** [CCG Parsing Algorithm with Incremental Tree Rotation](#)
Conference of the North American Chapter of the Association for Computational Linguistics
Miloš Stanojević and Mark Steedman
- CMCL 2019** [The active-filler strategy in a move-eager left-corner Minimalist Grammar parser](#)
Workshop on Cognitive Modeling and Computational Linguistics
 Tim Hunter, **Miloš Stanojević** and Edward Stabler
- CogACLL 2018** [A Sound and Complete Left-Corner Parsing for Minimalist Grammars](#)
Workshop on Cognitive Aspects of Computational Language Learning and Processing
Miloš Stanojević and Edward Stabler
- PROPOR 2018** [SICK-BR: a Portuguese corpus for inference](#)
International Conference on Computational Processing of the Portuguese Language
 Livy Real, Ana Rodrigues, Andressa Vieira e Silva, Beatriz Albiero, Bruna Thalenberg, Bruno Guide, Cindy Silva, Guilherme de Oliveira Lima, Igor CS Câmara, **Miloš Stanojević**, Rodrigo Souza, Valeria de Paiva
- PhD Thesis 2017** [Permutation Forests for Modeling Word Order in Machine Translation](#)
University of Amsterdam 2017
Miloš Stanojević
- EMNLP 2017** [Neural Discontinuous Constituency Parsing](#)
Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević and Raquel G. Alhama
- ACL 2017 short paper** [Alternative Objective Functions for Training MT Evaluation Metrics](#)
Annual Meeting of the Association for Computational Linguistics
Miloš Stanojević and Khalil Sima'an



- LACL 2016 [Minimalist Grammar Transition-Based Parsing](#)
Logical Aspects of Computational Linguistics
Miloš Stanojević
- COLING 2016 [Hierarchical Permutation Complexity for Word Order Evaluation](#)
International Conference on Computational Linguistics
Miloš Stanojević and Khalil Sima'an
- COLING 2016 [Universal Reordering via Linguistic Typology](#)
International Conference on Computational Linguistics
Joachim Daiber, **Miloš Stanojević** and Khalil Sima'an
- WMT 2016 [Examining the Relationship between Preordering and Word Order Freedom in Machine Translation](#)
Conference on Machine Translation
Joachim Daiber, **Miloš Stanojević**, Wilker Aziz and Khalil Sima'an
- WMT 2016 [Results of the WMT16 Metrics Shared Task](#)
Conference on Machine Translation
Ondřej Bojar, Yvette Graham, Amir Kamran and **Miloš Stanojević**
- WMT 2016 [Results of the WMT16 Tuning Shared Task](#)
Conference on Machine Translation
Bushra Jawaid, Amir Kamran, **Miloš Stanojević** and Ondřej Bojar
- EMNLP 2015 [Reordering Grammar Induction](#)
Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević and Khalil Sima'an
- PBML 2015 [Evaluating MT systems with BEER](#)
The Prague Bulletin of Mathematical Linguistics
Miloš Stanojević and Khalil Sima'an
- WMT 2015 [BEER 1.1: ILLC UvA submission to metrics and tuning task](#)
Workshop on Machine Translation
Miloš Stanojević and Khalil Sima'an
- WMT 2015 [Results of the WMT15 Metrics Shared Task](#)
Workshop on Machine Translation
Miloš Stanojević, Amir Kamran, Philipp Koehn and Ondřej Bojar
- WMT 2015 [Results of the WMT15 Tuning Shared Task](#)
Workshop on Machine Translation
Miloš Stanojević, Amir Kamran and Ondřej Bojar



SSST 2014	<u>Evaluating Word Order Recursively over Permutation-Forests</u> <i>Workshop on Syntax, Semantics and Structure in Statistical Translation</i> Miloš Stanojević and Khalil Sima'an
WMT 2014	<u>BEER: BEtter Evaluation as Ranking</u> <i>Workshop on Machine Translation</i> Miloš Stanojević and Khalil Sima'an
EMNLP 2014 short paper	<u>Fitting Sentence Level Translation Evaluation with Many Dense Features</u> <i>Conference on Empirical Methods in Natural Language Processing</i> Miloš Stanojević and Khalil Sima'an
WMT 2012	<u>Selecting data for English-to-Czech machine translation</u> <i>Workshop on Machine Translation</i> Aleš Tamchyna, Petra Galuščáková, Amir Kamran, Miloš Stanojević , Ondřej Bojar

CONTRIBUTIONS TO THE RESEARCH COMMUNITY

ARR Area Chair:	- ACL, NAACL, EMNLP, EACL, CoNLL conferences
Standing Reviewer:	- TACL journal (Impact Factor 6.9) - Computational Linguistics journal (Impact Factor 5.3)
Reviewer:	- Natural Language Processing Journal (NLP) (Impact Factor 1.9) - Language Resources and Evaluation Journal (LREC) (Impact Factor 1.8) - Journal of Language Modeling (Impact Factor not available)
Shared tasks organization:	- Workshop on Machine Translation (WMT) - Metrics task in 2015 & 2016 - Workshop on Machine Translation (WMT) - Tuning task in 2015 & 2016

SUPERVISION AND MENTORSHIP

<i>Johns Hopkins University</i>	2026- Alan Zhou (part of PhD dissertation committee)
<i>UCL supervision</i>	2026- Francesco Cipriani (PhD 2 nd co-supervisor) + 8 MA students in Computational Linguistics specialization + 2 BA student in General Linguistics
<i>UCL mentorship</i>	2026 Tal Nahari now Postdoctoral Fellow, Affective Brain Lab, UCL 2025 Maximilian Moser



now Research at King's College London

**Google DeepMind
internship mentor of PhD
students**

2026 **Yifu Qiu**
now PhD at University of Edinburgh

2025 **Zifan Jiang**
now PhD at University of Zurich

2024 **Matthias Lindemann**
now PostDoc at Instituto de Telecomunicações, Lisbon, Portugal

**Google DeepMind
scholar mentor**

2026 **Sindhu B. Hegde**
now PhD student at University of Oxford

2023 **Catherine Nana Nyaah Essuman**
now PhD student at Ontario Tech University

2022 **Mukhtar Mohamed**
now intern at Google DeepMind

**University of Edinburgh
supervision**

2019 **John Torr** (PhD Dissertation 2nd supervisor in the last year)
now RS at Apple Research, Cambridge

2021 **Seth Aycock** (MSc Dissertation main supervisor)
now PhD student at University of Amsterdam

2019 **Chiehmin Wei** (BA project supervisor)
now software engineer at Roblox

TEACHING

2025/2026
UCL

Lectures on Language Modeling within
PLIN0072 Advanced Computational Linguistics

2025/2026
UCL

Lectures on Parsing within
PLIN0034 Introduction to Computational Linguistics

2024/2025
UCL

Lectures on Parsing within
PLIN0034 Introduction to Computational Linguistics

2024/2025
UCL

Present linguistics students with career opportunities in the industry
UCL LingSoc Careers Event

2012/2013
University of Amsterdam

Statistical Structure in NLP

2013/2014
University of Amsterdam

Elements of Language Processing



2014/2015 <i>University of Amsterdam</i>	short course on Monte Carlo Inference for PCFG
2015/2016 <i>University of Amsterdam</i>	Natural Language Processing 2
2016/2017 <i>University of Amsterdam</i>	Natural Language Processing 2

INVITED TALKS

<i>LSSC keynote speaker</i> <i>Novi Sad, Serbia, 2026</i>	Interaction of Linguistics and Computer Science Then and Now Keynote speaker at conference <i>Language & Social Sciences in Contact: Artificial Intelligence (AI)</i>
<i>FlaNN Seminar</i> <i>Online, 2026</i>	Modeling Sentence Processing Difficulty with Bayesian Syntax <i>Formal Languages and Neural Networks Seminar</i>
<i>Stony Brook University</i> <i>New York, USA, 2025</i>	Incremental Sentence Processing of Right-Adjunction
<i>ILFC Seminar – CNRS,</i> <i>Paris, France, 2025</i>	Linguistic Universals in Grammars and Language Models
<i>Johns Hopkins University</i> <i>Baltimore, USA, 2024</i>	Syntactic Universals in Grammars and LLMs
<i>University of Amsterdam</i> <i>Netherlands, 2023</i>	Syntactic Structures and Large Language Models
<i>University of Cambridge</i> <i>Cambridge, UK, 2023</i>	The role of syntax in the world of large language models
<i>UCL</i> <i>London, UK, 2023</i>	What is the relation of Syntactic Parsing, Language Universals, Brain Activity and Large Language Models?
<i>ETH Zürich</i> <i>Zürich, Switzerland, 2022</i>	Testing Syntactic Theory on all Marr’s Levels of Analysis
<i>Mathematics of Language</i> <i>Montpellier, France, 2021</i>	CCG Combinators in Brain Activity and Linguistic Universals
<i>DeepMind</i> <i>London, UK, 2021</i>	Integrating Deep Learning with Deep Syntax
<i>Nuance NL & AI lab</i> <i>Sunnyvale, USA, 2017</i>	Learning Reordering Grammars



Rich Parsing Workshop
Amsterdam, 2016

Minimalist Grammar Transition-Based Parsing

MT Marathon
Prague, Czechia, 2015

Discriminative Training for Statistical Machine Translation

GRANTS

- 2026 **Capital Equipment Investment Fund by UKRI**
£185,025 for computing equipment
Miloš Stanojević (lead applicant) and Mario Giulianelli (co-applicant)
- 2025 **Language Sciences Incubator Fund by University of Cambridge**
£5,000 for research project “Improving Language Equity and Inclusion through AI”
lead applicants: Songqiao Xie and Xinxin Yan
co-applicants: Mario Giulianelli, Napoleon Katsos, Miloš Stanojević

SCHOLARSHIPS

- 2010 – 2012 **Erasmus Mundus scholarship for EM-LCT Masters Program**
- 2004 – 2010 **Scholarship from the Ministry of Science and Technological Development of Serbia**
awarded for taking part in international mathematical competitions
- 2002 – 2004 **Scholarship from the Ministry of Education of Serbia**
awarded for taking part in international mathematical competition
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